

# Family Movie Ratings Analysis 3a: Back To The Future

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# Introduction

This analysis examines movie ratings from our family for 6 recent popular movies. The participants are:

- **Randy** (56 Father)
- **Elle** (42, Mother)
- **Ariah** (9, Daughter)
- **Alexandra** (6, Daughter)
- **Don** (82, Grandfather)

The movies rated include both focus films (Barbie and KPop Demon Hunters) plus 4 additional recent popular movies suitable for a wide age range. In addition, we will attempt to use Global Baseline Estimates about the shows and the viewers.

## Data Loading

### Load Raw Data from Database

```
# Get all ratings with participant and movie details
ratings_data <- get_ratings_with_names(con)

# Display the dataset
knitr::kable(ratings_data,
              caption = "Complete Movie Ratings Dataset",
              col.names = c("Name", "Age", "Relationship", "Movie", "Year", "Genre", "Rating", "Notes"))
```

Table 1: Complete Movie Ratings Dataset

| Name      | Age | Relationship | Movie                          | Year | Genre            | Rating | Notes                               |
|-----------|-----|--------------|--------------------------------|------|------------------|--------|-------------------------------------|
| Don       | 82  | Grandfather  | Barbie                         | 2023 | Comedy/Fantasy   | NA     |                                     |
| Randy     | 45  | Father       | Barbie                         | 2023 | Comedy/Fantasy   | 3      | Funny but not really my thing       |
| Elle      | 42  | Mother       | Barbie                         | 2023 | Comedy/Fantasy   | 5      | Loved the humor and message!        |
| Ariah     | 9   | Daughter     | Barbie                         | 2023 | Comedy/Fantasy   | 4      | Pink and fun, good movie            |
| Alexandra | 6   | Daughter     | Barbie                         | 2023 | Comedy/Fantasy   | 5      | So pretty! I want to be Barbie      |
| Don       | 82  | Grandfather  | Elemental                      | 2023 | Animation/Family | 4      | Nice story, easy to follow          |
| Randy     | 45  | Father       | Elemental                      | 2023 | Animation/Family | 4      | Clever concept and execution        |
| Elle      | 42  | Mother       | Elemental                      | 2023 | Animation/Family | 5      | Beautiful message about differences |
| Ariah     | 9   | Daughter     | Elemental                      | 2023 | Animation/Family | 5      | Ember and Wade are so cute together |
| Alexandra | 6   | Daughter     | Elemental                      | 2023 | Animation/Family | 4      | Pretty fire and water characters    |
| Don       | 82  | Grandfather  | Guardians of the Galaxy Vol. 3 | 2023 | Action/Adventure | 2      | Too much action for me              |

| Name      | Age | Relationship | Movie                               | Year | Genre            | Rating | Notes                            |
|-----------|-----|--------------|-------------------------------------|------|------------------|--------|----------------------------------|
| Randy     | 45  | Father       | Guardians of the Galaxy Vol. 3      | 2023 | Action/Adventure | 4      | Good conclusion to the trilogy   |
| Elle      | 42  | Mother       | Guardians of the Galaxy Vol. 3      | 2023 | Action/Adventure | 3      | Bit too intense in parts         |
| Ariah     | 9   | Daughter     | Guardians of the Galaxy Vol. 3      | 2023 | Action/Adventure | 3      | Good but sad parts made me cry   |
| Alexandra | 6   | Daughter     | Guardians of the Galaxy Vol. 3      | 2023 | Action/Adventure | NA     | Too scary, did not finish        |
| Don       | 82  | Grandfather  | KPop Demon Hunters                  | 2023 | Action/Comedy    | NA     |                                  |
| Randy     | 45  | Father       | KPop Demon Hunters                  | 2023 | Action/Comedy    | 4      | Great action sequences           |
| Elle      | 42  | Mother       | KPop Demon Hunters                  | 2023 | Action/Comedy    | 4      | Surprisingly entertaining        |
| Ariah     | 9   | Daughter     | KPop Demon Hunters                  | 2023 | Action/Comedy    | 5      | Love the music and fighting      |
| Alexandra | 6   | Daughter     | KPop Demon Hunters                  | 2023 | Action/Comedy    | 3      | Music was good but scary parts   |
| Don       | 82  | Grandfather  | Spider-Man: Across the Spider-Verse | 2023 | Animation/Action | 3      | Good story, nice animation       |
| Randy     | 45  | Father       | Spider-Man: Across the Spider-Verse | 2023 | Animation/Action | 5      | Amazing animation and story      |
| Elle      | 42  | Mother       | Spider-Man: Across the Spider-Verse | 2023 | Animation/Action | 4      | Visually stunning                |
| Ariah     | 9   | Daughter     | Spider-Man: Across the Spider-Verse | 2023 | Animation/Action | 5      | Best superhero movie ever!       |
| Alexandra | 6   | Daughter     | Spider-Man: Across the Spider-Verse | 2023 | Animation/Action | 4      | Pretty colors but long           |
| Don       | 82  | Grandfather  | The Little Mermaid                  | 2023 | Musical/Fantasy  | 4      | Classic story, well done         |
| Randy     | 45  | Father       | The Little Mermaid                  | 2023 | Musical/Fantasy  | 3      | Good but prefer the original     |
| Elle      | 42  | Mother       | The Little Mermaid                  | 2023 | Musical/Fantasy  | 4      | Beautiful songs and visuals      |
| Ariah     | 9   | Daughter     | The Little Mermaid                  | 2023 | Musical/Fantasy  | 4      | Ariel is brave and inspiring     |
| Alexandra | 6   | Daughter     | The Little Mermaid                  | 2023 | Musical/Fantasy  | 5      | I want to be a mermaid princess! |

## Summary Statistics

```
# Get movie averages
movie_averages <- get_movie_averages(con)

knitr::kable(movie_averages,
  caption = "Movie Rating Averages",
  col.names = c("Movie Title", "Number of Ratings", "Average Rating", "Min Rating", "Max Rating")
)
```

Table 2: Movie Rating Averages

| Movie Title                         | Number of Ratings | Average Rating | Min Rating | Max Rating |
|-------------------------------------|-------------------|----------------|------------|------------|
| Elemental                           | 5                 | 4.40           | 4          | 5          |
| Barbie                              | 4                 | 4.25           | 3          | 5          |
| Spider-Man: Across the Spider-Verse | 5                 | 4.20           | 3          | 5          |
| KPop Demon Hunters                  | 4                 | 4.00           | 3          | 5          |
| The Little Mermaid                  | 5                 | 4.00           | 3          | 5          |
| Guardians of the Galaxy Vol. 3      | 4                 | 3.00           | 2          | 4          |

```
# Overall statistics
```

```
cat("Dataset Summary:\n")
```

```
## Dataset Summary:
```

```
cat("Total ratings collected:", nrow(ratings_data), "\n")
```

```
## Total ratings collected: 30
```

```
cat("Number of movies:", length(unique(ratings_data$movie_title)), "\n")
```

```
## Number of movies: 6
```

```
cat("Number of participants:", length(unique(ratings_data$person_name)), "\n")
```

```
## Number of participants: 5
```

```
cat("Average rating across all movies:", round(mean(ratings_data$rating), 2), "\n")
```

```
## Average rating across all movies: NA
```

## Data Analysis

### Rating Distribution by Movie

```
# Create bar chart of average ratings by movie
```

```
movie_avg_plot <- movie_averages %>%
```

```
  ggplot(aes(x = reorder(title, avg_rating), y = avg_rating, fill = title)) +
```

```
  geom_col(show.legend = FALSE) +
```

```
  geom_text(aes(label = avg_rating), hjust = -0.1) +
```

```
  coord_flip() +
```

```
  labs(title = "Average Movie Ratings",
```

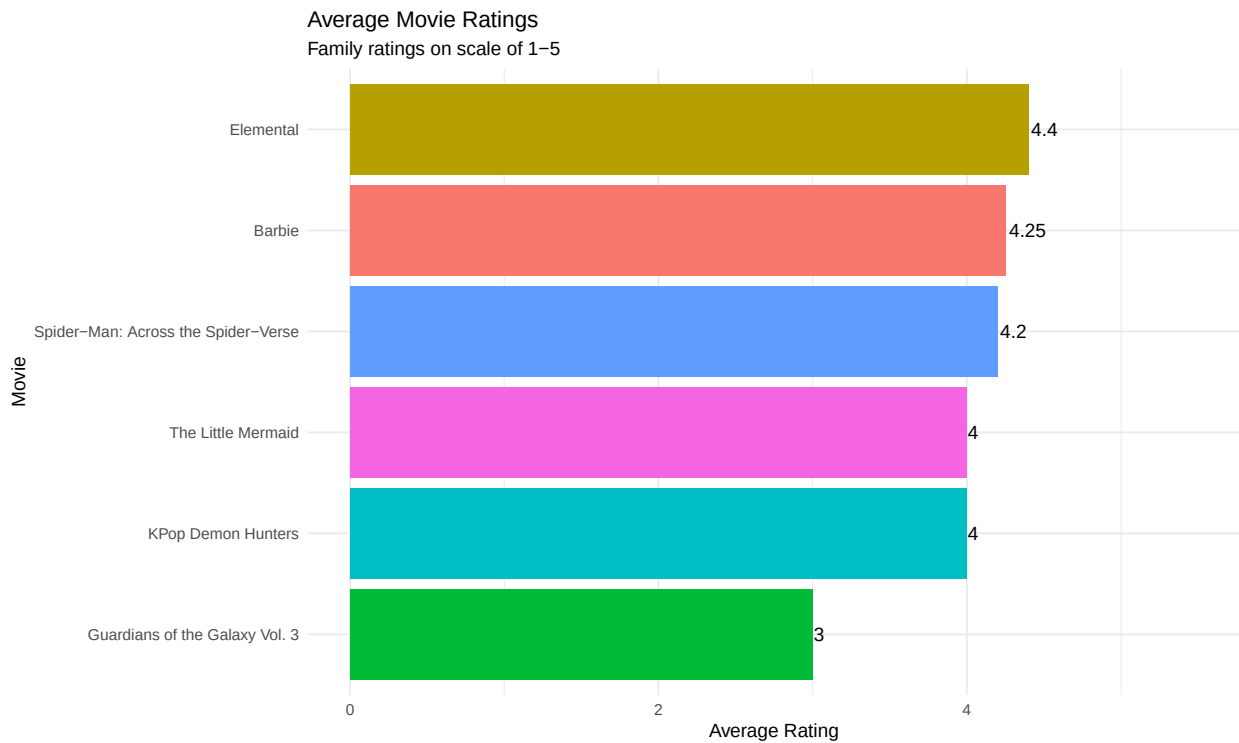
```
        subtitle = "Family ratings on scale of 1-5",
```

```
        x = "Movie",
```

```
        y = "Average Rating") +
```

```
theme_minimal() +
scale_y_continuous(limits = c(0, 5.5))

print(movie_avg_plot)
```



## Ratings by Age Group

```
# Create age groups for analysis
ratings_with_age_groups <- ratings_data %>%
  mutate(age_group = case_when(
    age <= 10 ~ "Children (6-9)",
    age <= 50 ~ "Adults (42-45)",
    age > 50 ~ "Senior (82+)"
  ))

# Average rating by age group
age_group_summary <- ratings_with_age_groups %>%
  group_by(age_group) %>%
  summarise(
    avg_rating = round(mean(rating), 2),
    count = n(),
    .groups = 'drop'
  )

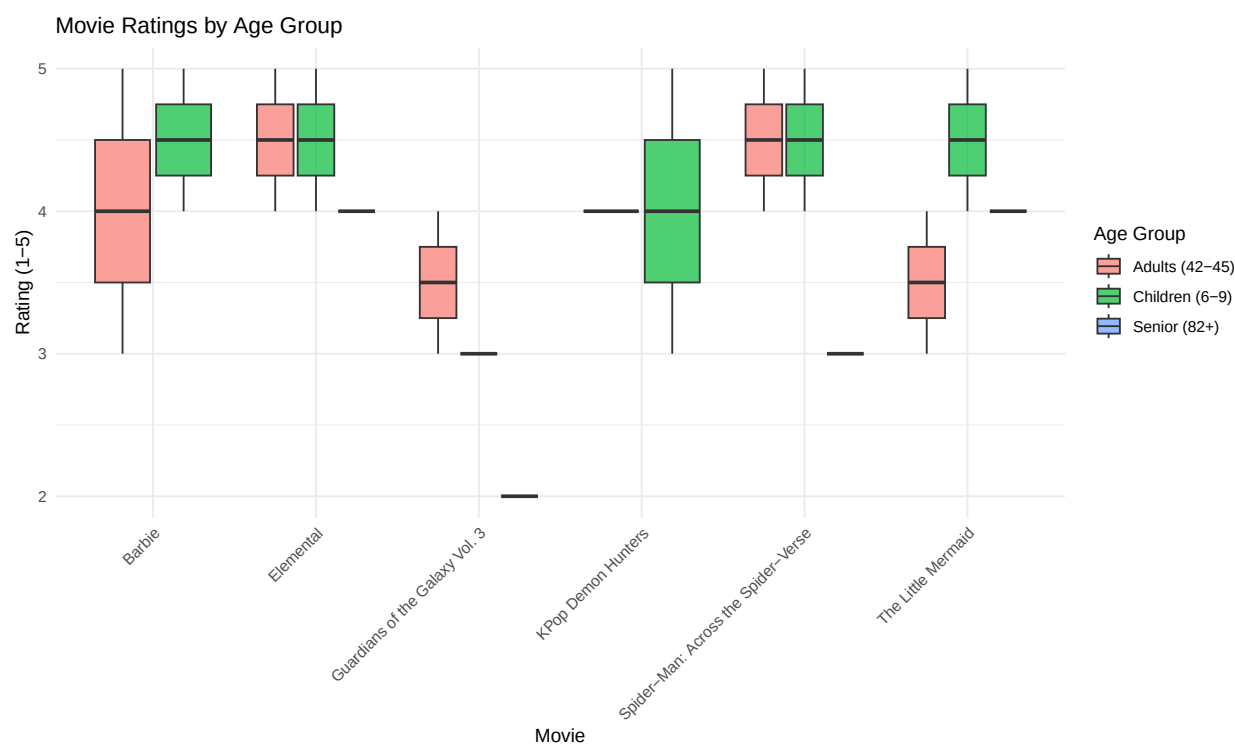
knitr::kable(age_group_summary,
  caption = "Average Ratings by Age Group")
```

Table 3: Average Ratings by Age Group

| age_group      | avg_rating | count |
|----------------|------------|-------|
| Adults (42-45) | 4          | 12    |
| Children (6-9) | NA         | 12    |
| Senior (82+)   | NA         | 6     |

```
# Plot ratings by age group and movie
age_movie_plot <- ratings_with_age_groups %>%
  ggplot(aes(x = movie_title, y = rating, fill = age_group)) +
  geom_boxplot(alpha = 0.7) +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1)) +
  labs(title = "Movie Ratings by Age Group",
       x = "Movie",
       y = "Rating (1-5)",
       fill = "Age Group")

print(age_movie_plot)
```



## Individual Rating Patterns

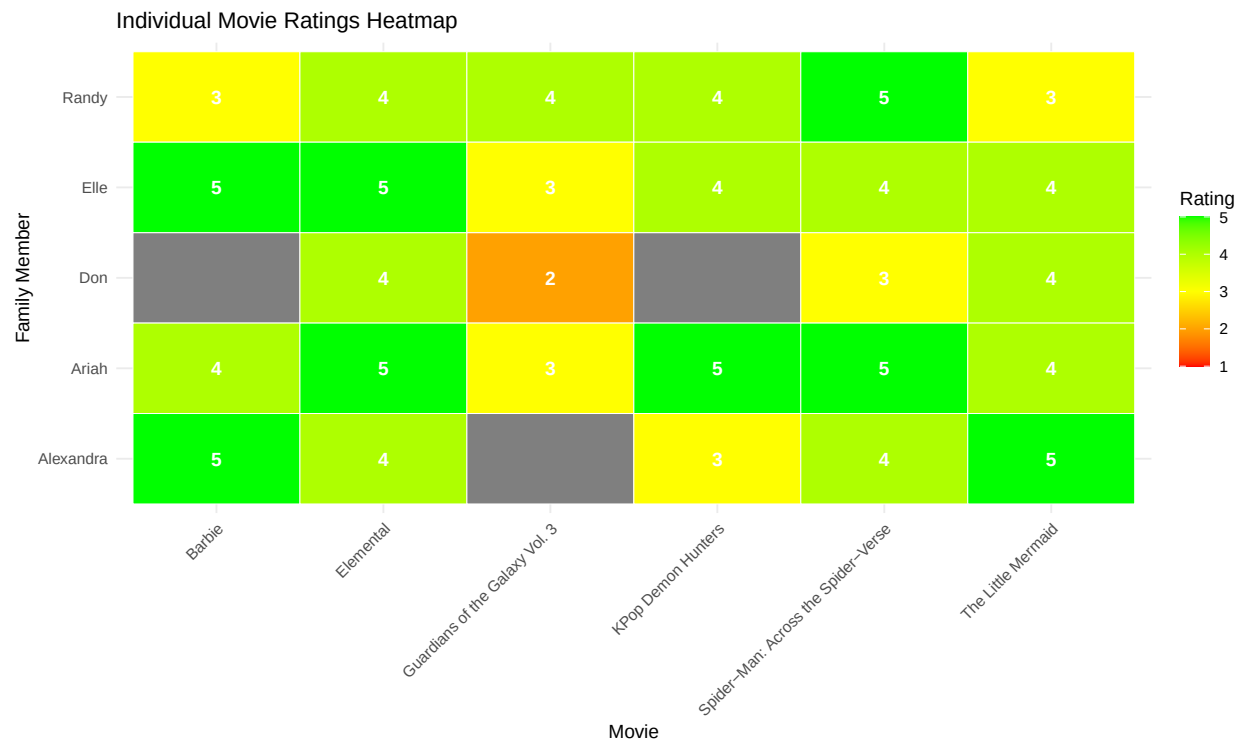
```
# Heatmap of individual ratings
individual_plot <- ratings_data %>%
  ggplot(aes(x = movie_title, y = person_name, fill = rating)) +
  geom_tile(color = "white") +
```

```

geom_text(aes(label = rating), color = "white", fontface = "bold") +
scale_fill_gradient2(low = "red", mid = "yellow", high = "green",
                     midpoint = 3, limits = c(1,5)) +
theme_minimal() +
theme(axis.text.x = element_text(angle = 45, hjust = 1)) +
labs(title = "Individual Movie Ratings Heatmap",
     x = "Movie",
     y = "Family Member",
     fill = "Rating")

print(individual_plot)

```



## Focus Movies Analysis

```

# Focus on the two focus movies
focus_movies <- ratings_data %>%
  filter(movie_title %in% c("Barbie", "KPop Demon Hunters"))

knitr::kable(focus_movies %>% select(person_name, age, movie_title, rating, notes),
             caption = "Ratings for focus Movies (Barbie & KPop Demon Hunters)")

```

Table 4: Ratings for focus Movies (Barbie & KPop Demon Hunters)

| person_name | age | movie_title | rating | notes |
|-------------|-----|-------------|--------|-------|
| Don         | 82  | Barbie      | NA     |       |

| person_name | age | movie_title        | rating | notes                          |
|-------------|-----|--------------------|--------|--------------------------------|
| Randy       | 45  | Barbie             | 3      | Funny but not really my thing  |
| Elle        | 42  | Barbie             | 5      | Loved the humor and message!   |
| Ariah       | 9   | Barbie             | 4      | Pink and fun, good movie       |
| Alexandra   | 6   | Barbie             | 5      | So pretty! I want to be Barbie |
| Don         | 82  | KPop Demon Hunters | NA     |                                |
| Randy       | 45  | KPop Demon Hunters | 4      | Great action sequences         |
| Elle        | 42  | KPop Demon Hunters | 4      | Surprisingly entertaining      |
| Ariah       | 9   | KPop Demon Hunters | 5      | Love the music and fighting    |
| Alexandra   | 6   | KPop Demon Hunters | 3      | Music was good but scary parts |

```
# Compare the focus movies
focus_comparison <- focus_movies %>%
  group_by(movie_title) %>%
  summarise(
    avg_rating = round(mean(rating), 2),
    ratings_range = paste(min(rating), "to", max(rating)),
    .groups = 'drop'
  )

knitr::kable(focus_comparison,
  caption = "focus Movies Comparison")
```

Table 5: focus Movies Comparison

| movie_title        | avg_rating | ratings_range |
|--------------------|------------|---------------|
| Barbie             | NA         | NA to NA      |
| KPop Demon Hunters | NA         | NA to NA      |

## Key Findings

```
# Find highest and lowest rated movies
best_movie <- movie_averages$title[which.max(movie_averages$avg_rating)]
worst_movie <- movie_averages$title[which.min(movie_averages$avg_rating)]

# Find most generous and harsh raters
person_averages <- ratings_data %>%
  group_by(person_name, age) %>%
  summarise(avg_rating = round(mean(rating), 2), .groups = 'drop') %>%
  arrange(desc(avg_rating))

most_generous <- person_averages$person_name[1]
most_critical <- person_averages$person_name[nrow(person_averages)]

cat("Key Findings:\n")
```

```
## Key Findings:
```

```
cat("• Highest rated movie:", best_movie,
    "(", max(movie_averages$avg_rating), "average )\n")
```

```
## • Highest rated movie: Elemental ( 4.4 average )
```

```
cat("• Lowest rated movie:", worst_movie,
    "(", min(movie_averages$avg_rating), "average )\n")
```

```
## • Lowest rated movie: Guardians of the Galaxy Vol. 3 ( 3 average )
```

```
cat("• Most generous rater:", most_generous,
    "(", max(person_averages$avg_rating), "average )\n")
```

```
## • Most generous rater: Arianne ( NA average )
```

```
cat("• Most critical rater:", most_critical,
    "(", min(person_averages$avg_rating), "average )\n")
```

```
## • Most critical rater: Don ( NA average )
```

## Global Baseline Estimate Recommendation System

### Overview

The Global Baseline Estimate is a non-personalized recommendation algorithm that predicts ratings based on:

- The overall average rating across all movies and users
- How each user tends to rate compared to the average (user bias)
- How each movie tends to be rated compared to the average (movie bias)

### Implementation

```
# Function to implement Global Baseline Estimate
calculate_global_baseline <- function(ratings_data) {
  # Calculate overall mean rating
  mean_all_ratings <- mean(ratings_data$rating, na.rm = TRUE)

  # Calculate user statistics and biases
  user_stats <- ratings_data %>%
    group_by(person_name) %>%
    summarise(
      user_avg = mean(rating, na.rm = TRUE),
      n_ratings = n(),
      .groups = 'drop'
    ) %>%
    mutate(user_bias = user_avg - mean_all_ratings)

  # Calculate movie statistics and biases
  movie_stats <- ratings_data %>%
```

```

    group_by(movie_title) %>%
    summarise(
      movie_avg = mean(rating, na.rm = TRUE),
      n_ratings = n(),
      .groups = 'drop'
    ) %>%
    mutate(movie_bias = movie_avg - mean_all_ratings)

return(list(
  mean_all = mean_all_ratings,
  user_stats = user_stats,
  movie_stats = movie_stats
))
}

# Function to predict a rating
predict_rating <- function(user, movie, baseline_model) {
  user_bias <- baseline_model$user_stats %>%
    filter(person_name == user) %>%
    pull(user_bias)

  movie_bias <- baseline_model$movie_stats %>%
    filter(movie_title == movie) %>%
    pull(movie_bias)

  # Handle missing values
  if (length(user_bias) == 0) user_bias <- 0
  if (length(movie_bias) == 0) movie_bias <- 0

  # Calculate prediction
  predicted <- baseline_model$mean_all + user_bias + movie_bias

  # Constrain to 1-5 scale
  return(max(1, min(5, predicted)))
}

```

## Model Training

```

# Train the model
baseline_model <- calculate_global_baseline(ratings_data)

# Display overall statistics
cat("Global Mean Rating:", round(baseline_model$mean_all, 2), "\n\n")

```

```
## Global Mean Rating: 4
```

```

# Display user biases
knitr::kable(baseline_model$user_stats %>%
  arrange(desc(user_bias)) %>%
  mutate(across(where(is.numeric), ~round(., 2))),
  caption = "User Rating Patterns and Biases")

```

Table 6: User Rating Patterns and Biases

| person_name | user_avg | n_ratings | user_bias |
|-------------|----------|-----------|-----------|
| Ariah       | 4.33     | 6         | 0.33      |
| Alexandra   | 4.20     | 6         | 0.20      |
| Elle        | 4.17     | 6         | 0.17      |
| Randy       | 3.83     | 6         | -0.17     |
| Don         | 3.25     | 6         | -0.75     |

```
# Display movie biases
knitr::kable(baseline_model$movie_stats %>%
  arrange(desc(movie_bias)) %>%
  mutate(across(where(is.numeric), ~round(., 2))),
  caption = "Movie Rating Patterns and Biases")
```

Table 7: Movie Rating Patterns and Biases

| movie_title                         | movie_avg | n_ratings | movie_bias |
|-------------------------------------|-----------|-----------|------------|
| Elemental                           | 4.40      | 5         | 0.40       |
| Barbie                              | 4.25      | 5         | 0.25       |
| Spider-Man: Across the Spider-Verse | 4.20      | 5         | 0.20       |
| KPop Demon Hunters                  | 4.00      | 5         | 0.00       |
| The Little Mermaid                  | 4.00      | 5         | 0.00       |
| Guardians of the Galaxy Vol. 3      | 3.00      | 5         | -1.00      |

## Generating Predictions

```
# Create all possible user-movie combinations
all_combinations <- expand_grid(
  person_name = unique(ratings_data$person_name),
  movie_title = unique(ratings_data$movie_title),
  stringsAsFactors = FALSE
)

# Add predictions
predictions <- all_combinations %>%
  mutate(
    predicted_rating = mapply(predict_rating, person_name, movie_title,
                              MoreArgs = list(baseline_model = baseline_model))
  ) %>%
  left_join(ratings_data %>% select(person_name, movie_title, actual_rating = rating),
    by = c("person_name", "movie_title"))

# Show sample predictions
sample_predictions <- predictions %>%
  filter(!is.na(actual_rating)) %>%
  sample_n(min(10, n())) %>%
  mutate(
```

```

    error = actual_rating - predicted_rating,
    abs_error = abs(error)
  ) %>%
  arrange(person_name, movie_title)

knitr::kable(sample_predictions %>%
  mutate(across(where(is.numeric), ~round(., 2))),
  caption = "Sample Predictions vs Actual Ratings")

```

Table 8: Sample Predictions vs Actual Ratings

| person_name | movie_title                         | predicted_rating | actual_rating | error | abs_error |
|-------------|-------------------------------------|------------------|---------------|-------|-----------|
| Alexandra   | Elemental                           | 4.60             | 4             | -0.60 | 0.60      |
| Alexandra   | Spider-Man: Across the Spider-Verse | 4.40             | 4             | -0.40 | 0.40      |
| Alexandra   | The Little Mermaid                  | 4.20             | 5             | 0.80  | 0.80      |
| Ariah       | Guardians of the Galaxy Vol. 3      | 3.33             | 3             | -0.33 | 0.33      |
| Ariah       | The Little Mermaid                  | 4.33             | 4             | -0.33 | 0.33      |
| Don         | The Little Mermaid                  | 3.25             | 4             | 0.75  | 0.75      |
| Elle        | Guardians of the Galaxy Vol. 3      | 3.17             | 3             | -0.17 | 0.17      |
| Elle        | Spider-Man: Across the Spider-Verse | 4.37             | 4             | -0.37 | 0.37      |
| Randy       | Spider-Man: Across the Spider-Verse | 4.03             | 5             | 0.97  | 0.97      |
| Randy       | The Little Mermaid                  | 3.83             | 3             | -0.83 | 0.83      |

## Model Performance

```

# Calculate performance metrics on known ratings
known_ratings <- predictions %>%
  filter(!is.na(actual_rating))

mae <- mean(abs(known_ratings$actual_rating - known_ratings$predicted_rating))
rmse <- sqrt(mean((known_ratings$actual_rating - known_ratings$predicted_rating)^2))

cat("Model Performance Metrics:\n")

```

```
## Model Performance Metrics:
```

```
cat("- Mean Absolute Error (MAE):", round(mae, 3), "\n")
```

```
## - Mean Absolute Error (MAE): 0.531
```

```
cat("- Root Mean Square Error (RMSE):", round(rmse, 3), "\n")
```

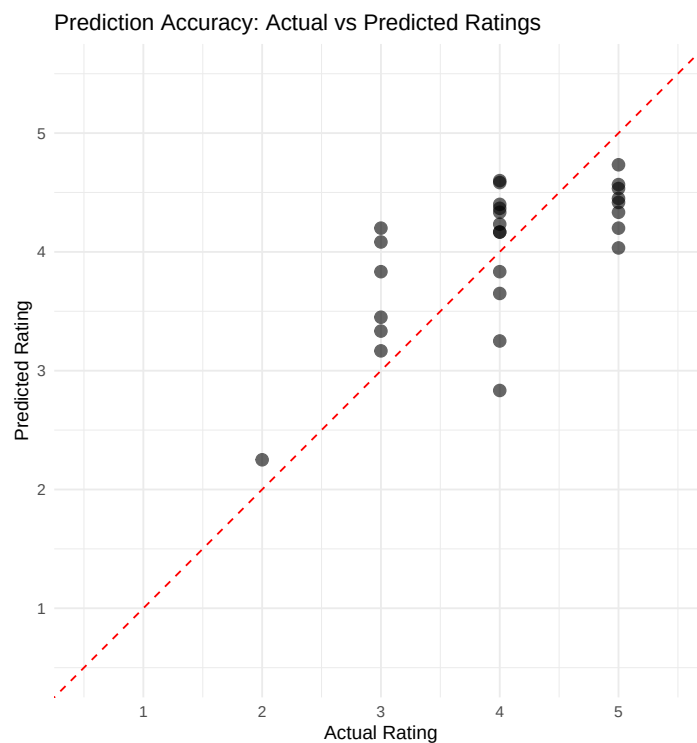
```
## - Root Mean Square Error (RMSE): 0.613
```

```
cat("- Number of predictions evaluated:", nrow(known_ratings), "\n")
```

```
## - Number of predictions evaluated: 27
```

```
# Visualize prediction accuracy
accuracy_plot <- known_ratings %>%
  ggplot(aes(x = actual_rating, y = predicted_rating)) +
  geom_point(alpha = 0.6, size = 3) +
  geom_abline(intercept = 0, slope = 1, linetype = "dashed", color = "red") +
  scale_x_continuous(breaks = 1:5, limits = c(0.5, 5.5)) +
  scale_y_continuous(breaks = 1:5, limits = c(0.5, 5.5)) +
  labs(title = "Prediction Accuracy: Actual vs Predicted Ratings",
       x = "Actual Rating",
       y = "Predicted Rating") +
  theme_minimal() +
  coord_equal()

print(accuracy_plot)
```



## Recommendations

```
# Generate recommendations for each user
recommendations <- predictions %>%
  filter(is.na(actual_rating)) %>%
  group_by(person_name) %>%
```

```

arrange(desc(predicted_rating)) %>%
slice_head(n = 2) %>%
select(person_name, movie_title, predicted_rating) %>%
mutate(predicted_rating = round(predicted_rating, 2))

knitr::kable(recommendations,
             caption = "Top 2 Movie Recommendations for Each Family Member")

```

Table 9: Top 2 Movie Recommendations for Each Family Member

| person_name | movie_title                    | predicted_rating |
|-------------|--------------------------------|------------------|
| Alexandra   | Guardians of the Galaxy Vol. 3 | 3.20             |
| Don         | Barbie                         | 3.50             |
| Don         | KPop Demon Hunters             | 3.25             |

```

# Focus on specific example (similar to Excel spreadsheet)
# Let's predict what Don would rate Barbie
example_user <- "Don"
example_movie <- "Barbie"

if (example_user %in% ratings_data$person_name &&
    example_movie %in% ratings_data$movie_title) {

  user_bias <- baseline_model$user_stats %>%
    filter(person_name == example_user) %>%
    pull(user_bias)

  movie_bias <- baseline_model$movie_stats %>%
    filter(movie_title == example_movie) %>%
    pull(movie_bias)

  predicted <- predict_rating(example_user, example_movie, baseline_model)

  cat("\n### Example Prediction Detail\n")
  cat("Predicting", example_user, "'s rating for", example_movie, "\n")
  cat("- Global mean rating:", round(baseline_model$mean_all, 2), "\n")
  cat("- ", example_user, "'s bias (tendency to rate higher/lower):", round(user_bias, 2), "\n")
  cat("- ", example_movie, "'s bias (tends to be rated higher/lower):", round(movie_bias, 2), "\n")
  cat("- Predicted rating:", round(baseline_model$mean_all, 2), "+",
        round(user_bias, 2), "+", round(movie_bias, 2), "=", round(predicted, 2), "\n")
}

```

```

##
## ### Example Prediction Detail
## Predicting Don 's rating for Barbie
## - Global mean rating: 4
## - Don 's bias (tendency to rate higher/lower): -0.75
## - Barbie 's bias (tends to be rated higher/lower): 0.25
## - Predicted rating: 4 + -0.75 + 0.25 = 3.5

```

## Recommendation-insights

```
# Analyze which movies are most recommended
most_recommended <- predictions %>%
  filter(is.na(actual_rating)) %>%
  group_by(movie_title) %>%
  summarise(
    avg_predicted = mean(predicted_rating),
    times_recommended = sum(predicted_rating >= 4),
    .groups = 'drop'
  ) %>%
  arrange(desc(avg_predicted))

knitr::kable(most_recommended %>%
  mutate(avg_predicted = round(avg_predicted, 2)),
  caption = "Movies Most Likely to be Enjoyed by Family Members Who Haven't Seen Them")
```

Table 10: Movies Most Likely to be Enjoyed by Family Members  
Who Haven't Seen Them

| movie_title                    | avg_predicted | times_recommended |
|--------------------------------|---------------|-------------------|
| Barbie                         | 3.50          | 0                 |
| KPop Demon Hunters             | 3.25          | 0                 |
| Guardians of the Galaxy Vol. 3 | 3.20          | 0                 |

## Technical Details

```
# Show database structure
tables <- dbListTables(con)
cat("DDEV Database tables:", paste(tables, collapse = ", "), "\n")
```

```
## DDEV Database tables: movies, ratings, people
```

```
# Show connection info
cat("\nDatabase connection details:\n")
```

```
##
## Database connection details:
```

```
cat("Host: 127.0.0.1\n")
```

```
## Host: 127.0.0.1
```

```
cat("Port: 3306\n")
```

```
## Port: 3306
```

```
cat("Database: db\n")
```

```
## Database: db
```

```
cat("User: db\n")
```

```
## User: db
```

## Methodology Notes

- **Data Collection:** Family members rated movies on a scale of 1-5
- **Storage:** Ratings stored in MariaDB database via DDEV
- **Analysis:** Loaded into R dataframes using RMariaDB package
- **Reproducibility:** Complete DDEV setup ensures consistent environment across platforms

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*Analysis completed on 2025-09-14 using R, MariaDB, and DDEV*